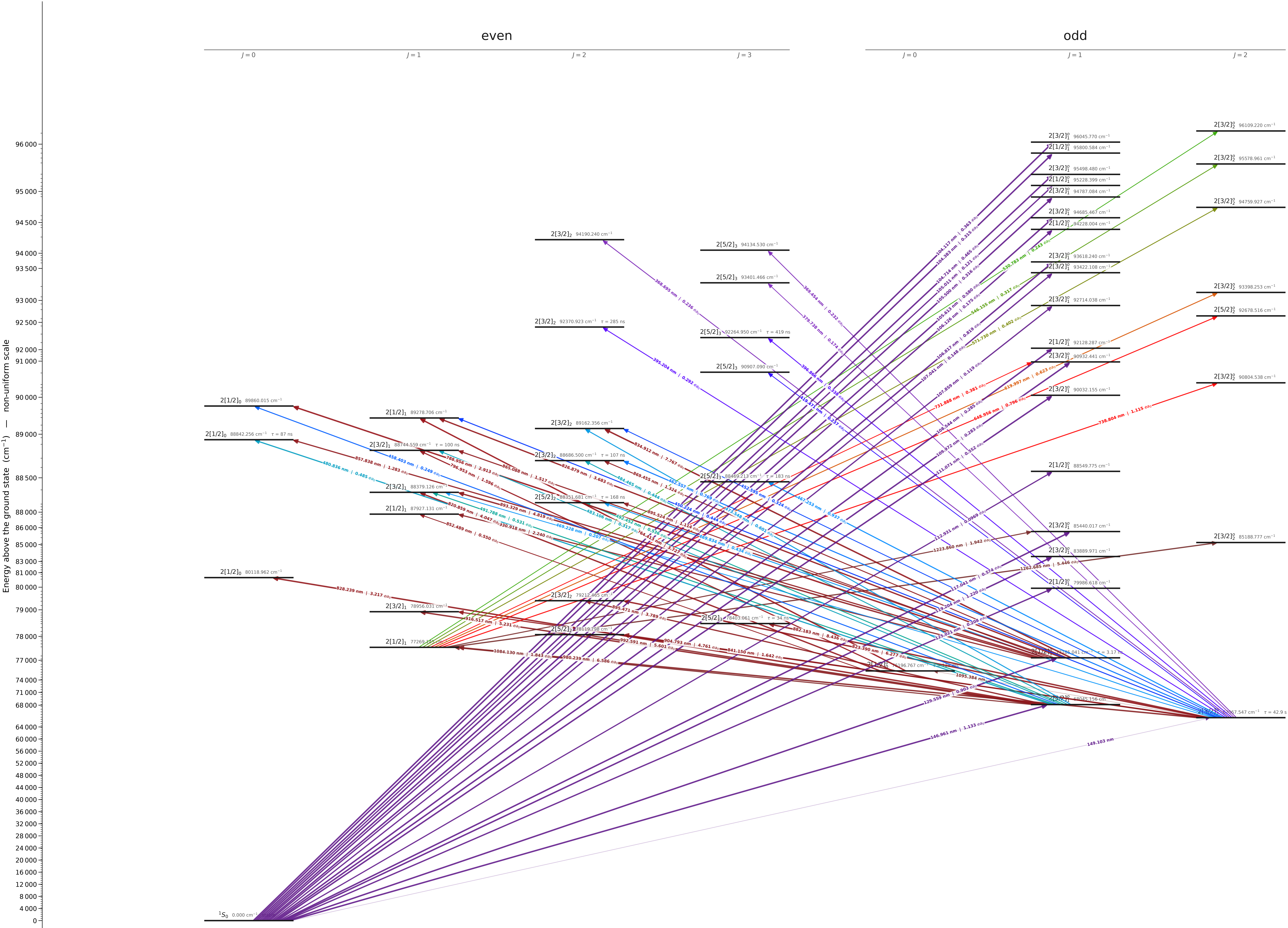
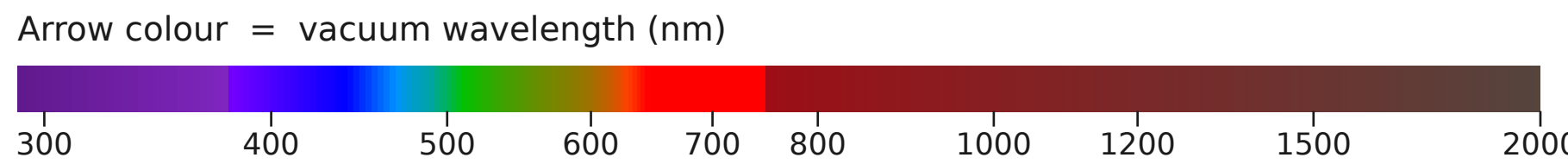


Xenon

Xe (Xe I). Strongest classified transitions below 2 μm (NIST): 51 levels, 72 lines.

Xe⁺ ionisation limit 97833.79 cm⁻¹ = 12.12984 eV (102.214 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||J\rangle$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U er J\rangle$ ($e a_0$)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$^1S_0 - 2[1/2]_1^o$	104.3835	-	2872.02930(10)	0.315	5.900e+07	-	-	104.4 nm VUV line: absolute calibration, hyperfine struct...
$^1S_0 - 2[3/2]_1^o$	105.6128	-	2838.5989	0.580	1.930e+08	-	-	105.6 nm VUV line: absolute calibration, hyperfine struct...
$^1S_0 - 2[1/2]_1^o$	106.1256	-	2824.8845	0.175	1.740e+07	-	-	106.1 nm VUV line: absolute calibration used for the ion...
$^1S_0 - 2[3/2]_1^o$	106.8168	-	2806.6042	0.819	3.720e+08	-	-	106.8 nm VUV line
$^1S_0 - 2[3/2]_1^o$	109.9718	-	2726.08612	0.283	4.080e+07	-	-	110 nm line measured by Ramsey-comb
$^1S_0 - 2[3/2]_1^o$	146.9612	-	2039.9425	1.133	2.730e+08	-	-	147 nm resonance line: VUV-lamp first step of the optical...
$^1S_0 - 2[3/2]_1^o$	149.1034	-	2010.6345	-	2.331e-02	3.71e-09	100	149.1 nm M2 decay of the metastable 3P2 state
$2[3/2]_2^o - 2[1/2]_1$	450.2241	450.0979	665.873836	0.444	1.460e+06	-	-	450.1 nm
$2[1/2]_1^o - 2[1/2]_1$	764.4127	764.2023	392.186651	3.727	2.100e+07	-	-	764.2 nm
$2[1/2]_1^o - 2[3/2]_1$	820.8591	820.6335	365.2179	4.047	2.000e+07	-	-	820.86 nm
$2[3/2]_2^o - 2[3/2]_1$	823.3897	823.1634	364.0955	6.277	2.860e+07	-	-	823.2 nm
$2[3/2]_2^o - 2[3/2]_1$	841.1501	840.9190	356.4078	1.642	3.060e+06	-	-	841.15 nm
$2[3/2]_2^o - 2[5/2]_3$	882.1832	881.9410	339.8302	8.436	3.000e+07	4.681	100	881.9 nm
$2[3/2]_1^o - 2[3/2]_2$	895.4709	895.2251	334.7875	3.789	8.100e+06	-	-	895.5 nm second step
$2[3/2]_1^o - 2[3/2]_1$	916.5168	916.2653	327.0998	5.231	2.400e+07	-	-	916 nm optical pumping into 6p[3/2]1, which decays to the...
$2[3/2]_2^o - 2[1/2]_1^o$	1095.3836	1095.0837	273.687184	-	-	1.243e-06	-	proposed optical frequency standard: two-photon clock tra...

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

¹²⁹Xe hyperfine structure (*I* = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$2[3/2]_2^o$	-2384.4(12)	-	F=3/2: +3576.60 F=5/2: -2384.40
$2[3/2]_1^o$	-959.1(7)	-	F=1/2: +959.10 F=3/2: -479.55
$2[1/2]_1^o$	-5808.0(20)	-	F=1/2: +5808.00 F=3/2: -2904.00
$2[3/2]_1$	-1308.9(6)	-	F=1/2: +1308.89 F=3/2: -654.45
$2[3/2]_2$	-889.6(4)	-	F=3/2: +1334.40 F=5/2: -889.60
$2[3/2]_1$	-4424.0(10)	-	F=1/2: +4424.00 F=3/2: -2212.00
$2[3/2]_1$	-2740.4(9)	-	F=1/2: +2740.40 F=3/2: -1370.20
$2[3/2]_2$	-2895(5)	-	F=3/2: +4341.90 F=5/2: -2894.60
$2[1/2]_1$	1977.0(10)	-	F=1/2: -1977.00 F=3/2: +988.50
$2[3/2]_2^o$	1657(12)	-	F=1/2: -1657.00 F=3/2: +828.50
$2[1/2]_1^o$	-579(6)	-	F=1/2: +579.00 F=3/2: -289.50
$2[3/2]_2^o$	913(23)	-	F=1/2: -913.00 F=3/2: +456.50
$2[1/2]_1^o$	-4548(15)	-	F=1/2: +4548.00 F=3/2: -2274.00

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^1S_0 - 2[1/2]_1^o$	104.383	136-134	213(12)
$^1S_0 - 2[1/2]_1^o$	104.383	136-132	365(18)
$^1S_0 - 2[1/2]_1^o$	104.383	136-131	492(20)
$^1S_0 - 2[1/2]_1^o$	104.383	136-130	523(24)
$^1S_0 - 2[1/2]_1^o$	104.383	136-129	678(24)
$^1S_0 - 2[1/2]_1^o$	104.383	136-128	692(22)
$^1S_0 - 2[1/2]_1^o$	104.383	136-126	869(25)
$^1S_0 - 2[1/2]_1^o$	104.383	136-124	1069(22)
$^1S_0 - 2[3/2]_1^o$	105.613	136-134	219(15)
$^1S_0 - 2[3/2]_1^o$	105.613	136-132	354(20)
$^1S_0 - 2[3/2]_1^o$	105.613	136-131	545(17)
$^1S_0 - 2[3/2]_1^o$	105.613	136-130	529(30)
$^1S_0 - 2[3/2]_1^o$	105.613	136-129	684(27)
$^1S_0 - 2[3/2]_1^o$	105.613	136-128	726(20)
$^1S_0 - 2[3/2]_1^o$	105.613	136-126	912(20)
$^1S_0 - 2[3/2]_1^o$	105.613	136-124	1116(20)
$^1S_0 - 2[1/2]_1^o$	106.126	136-134	207(12)
$^1S_0 - 2[1/2]_1^o$	106.126	136-132	384(18)
$^1S_0 - 2[1/2]_1^o$	106.126	136-131	531(23)
$^1S_0 - 2[1/2]_1^o$	106.126	136-130	557(11)
$^1S_0 - 2[1/2]_1^o$	106.126	136-129	675(14)
$^1S_0 - 2[1/2]_1^o$	106.126	136-128	747(19)

Shift = $\nu(\text{first isotope}) - \nu(\text{second isotope})$.